Active Inference Math Session ~ July 23, 2024

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and so my plan since we already have momentum here is we've been reviewing the a big idea of probability distributions and one example we can go to is what's called a a coin biased coin so not all coins have to be 50/50 there can be a particular probability that is not one half that the pro that the coin winds up on heads and this is a nice starter example for some of the concepts that we might want to look at and the first concept that we can look at for a single distribution is what's called Shannon entropy now if you look at the definition of Shannon entropy on Wikipedia which I will I will look at but not pull up the formula in is not too hard to calculate and talk about what happens here in just a second but the formula for the probability distribution um is we're going to sum over all of the possible events so X small X is a single event and big X is the sample space which is all possible events that could occur and we're going to add the we're going to do a minus in front I don't know why Shannon entropy is commonly written with h maybe someone else does but h of X big X X because it belongs to the space is negative the sum of over all possible events of the probability of each event times the log of the probability of each event so maybe this looks terrible symbolically maybe it doesn't um let's see what that looks like with a 50/50 coin so if we look at a 50/50 Fair coin then the probability of of H is 0.5 probability of tails is 0.5 this H and T are examples of the small X so our sample space our big X overall would be that we have two events heads or tails and each event has a probability such that the probabilities add up to one so if I want to calculate um it's less than ideal that H is reused here so maybe I'll make this be a script H to make it more distinct so if I want to look at the Shannon entropy of this distribution I'm just going to take the probab times so it's going to be minus as it starts with the minus it's going to be 0.5 Times log of 0.5 because I'm taking care of the H here and then I have to do the same thing for the T which is 0.5 and we can calculate that I don't know what log of 0.5 is off hand can anyone calculate this and tell me what the number is if not I can do it but yall are welcome to jump in just now it seems to be about negative 0.693 you're saying that's for the overall number oh sorry no just log of 0.5 I see okay yeah now I got you um and then it turns out miraculously that when you calculate because this is you know dist distribution you get

Negative log of 0.5 so this number is about 0693 and okay so that's a number and I guess the question is what does that tell us well what I feel might be helpful is to calculate this not for this single one but now for arbitrary probability so let's now say we look at any coin any coin and we're going to look at probability its heads and probab probability its tails so I'm going to use so I'm going to use X for the probability that we have heads and I don't know what x is X is a number it tells me how biased the coin is towards heads and then I'm going to look at the one minus t or 1 minus X which is the probability of Tails so this is a biased coin and how biased it is is determined by X and we can do the same thing we just did let's call this um Shannon entropy of our I'm going to use capital X subb and b stands for bias and because I decided to use B here I'm actually going to use B here because I I don't want to overload these X's especially when they tend to look so much alike so this will be probability of heads is B probability of tails is one minus B so now I have negative and I'm going to add up B [Music] * log of B+ 1us btimes log of 1us B now this depends on B so I can't come up with a single number here but I can make a plot of different things this could look like and let's see what it looks like e share the screen but not destroy the Whiteboard so let me see can I save this so now I'm going to look at the plot of this and I want to look at this from what's the smallest value that b could possibly be if B is the probability that the coin comes up heads well zero put that it could be zero and it could be at most one there's actually a little bit of weirdness here because we can't take log of zero I have to come back to that uh but I'm going to plot this and to see what it looks like unfortunately now I have to switch to X and probably get away with not doing that just make that menion okay and I need a negative here and you have to be careful about which log Bas we using to be consistent okay this is kind of interesting this is the graph of Shannon in P so the yes um would something so something at zero is actually a no no uh 0 times natural log of zero is undefined and uh there's a a convention that people make because in math how are things defined they're not defined you know they're defined by us essentially by social convention so natural log of zero is an undefined quantity um so what you do to work around that is you just Define that to be zero and you'll see people do this colon equals to say hey I am defining this to be this and it turns out it doesn't cause any problems uh when you're when you're really close to zero it's not a problem it's only a problem right at zero and that's why you see you see that already when you scroll in here close to b equals z so this is on the x- axis the probability that the coin comes up heads and then on the y- axis I have the Shannon entropy for that distribution of coin values so what do you see here is the probability of the coin being heads approaches 0. five or you know the the probability of the you know the distribution p and then one minus B the more similar they are the less information we have well the more similar they are the greater the entropy of the distribution and perhaps we can then interpret that as saying the less certain we are about uh what the uh what the bias of the coin is but then again we know the Bu The Coin so this is definitely a a maximum value at 0.5 and uh you said something about and it it's shaped like a parabola it's um oh concave down the entire way and you said can you okay you said something that had a lot in it and I if you'd be willing to just repeat it about um this interpretation of why the maximum Shannon entropy occurs when the probabilities are equal or what that what that just pops into your mind yeah well I I guess colloquially um you know entropy is is a measure of the spread of a probability distribution so visually if we were to plot say the probability distribution at uh P equal .5 we'd have two bars of height .5 and they'd be kind of exactly uh spread out you know whereas if we had p p equal like one then we'd have a massive bar at one and a tiny non-existent bar at zero and that would not be very you know quote unquote spread out so P equal 0.5 is where the distribution is is the most spread out so that that then corresponds to the the greatest entropy okay okay so the greatest entropy occurs and you're saying that we would have a a bar looks like this HT and each of them has a height of 0.5 the two heights are equal and that's what maximum intri looks like versus absolute certainty of um Tails let's say or yeah absolute certainty of Tails I think actually the way I set this up IE is backwards this is because I said that b was the probability of heads so let me be consistent to not confuse people um so then it's probability of heads is 1.0 okay um then that has zero entropy and you said that clo entropy is a spread of the distribution and that's I'm totally okay with that um I'd also say it's the uncertainty about individual events or uh again being collo colloquial How likely am I to be how good a strategy is guessing one of the values uh so if I look here where B is equal to 0.9 that corresponds to the probability of heads is 0.9 and the probability of t is 0.1 so I expect that if I say heads I'm going to be right most of the time in other words I feel I have a lot of information about what the next coin flip is likely to be and if I use a guessing strategy where I guess heads all the time then I'm probably going to be right most of the time right and so I I have a good approximation strategy does that make sense I hope it does I know that we're going through this slowly and carefully um but I feel it's good to build things so maybe we could look also at what the just the standard Wikipedia treatment of this is so what does it say about entropy I'm going to take a screenshot of this so it says in information Theory the entropy of a random variable is the average level of information surprise or uncertainty inherent to the variable's possible outcomes given a discrete random variable X and a discrete random variable X here is the same thing as what I was calling a discrete probability distribution which takes values in the set X that's sample space then we calculate the entropy like this we do have a choice in the base of the logarithm so if we use the natural log we get what are called Nets if we use the base two we get bits the motivation for this is coming from communication Theory Shannon was working on um creating signals to send be worked for telephone company he was working on creating uh yeah he was working on creating a way of compressing information to send it and then measuring how good a job had been done so there's actually something else U that pops up here that we used before or could have defined even before we jumped into entropy which was the information content which is the negative logarithm of the probability of an event or the surprisal it's also called so in entropy measures the entropy is a weighted average of surprisal entropy is an expected or average amount of information compay conveyed by identifying the outcome of a random trial it's how much information we're going to get every time we flip the coin so I want to go back and pull this into kback leer Divergence pack ler Divergence was originally called relative entropy and that's why it's important to have some sense of entropy before going into cobc webler because it's based on entropy let's take a moment and look at formula the just this has popped up in chapter this pops up in chapter two of the textbook here we're going to be given two probability distributions and what I'd like to do to make this very concrete is to imagine that we have two biased coins or maybe a bias coin and a Fair coin so I'm going to say let's just say p is a Fair coin 50% heads Q is a biased coin which has B percent heads probability of B of course you can do this for any distribution um any pair of probability distributions do you see how this looks like the formula for Shannon entropy if Q ofx were equal to one and this were made negative so the original formula was very similar to this but it had a negative in front of the summation and there was no Q of X in the denominator let's calculate the KL Divergence for two coins um so to do this I will need a biased coin would anyone be willing to create a biased coin for me which all you have to do is specify the probability of heads s okay why not what a great you said 7 yeah Okay cool so let's have P our first probability distribution be fair which is just heads and tails are both 50% and then my biased coin Q is going to be that probability of heads is 7 so now I'm going to calculate the Rel entropy or the kak leader Divergence of and the order does matter this is not symmetric so I'm going to go P first and I'm going to add up over my events the P of x time log of p ofx over Q ofx so this is going to be summing over two things one that corresponds to H and one that corresponds to T so we're going to have 0.5 so first let's work with the H part 0 0.5 * log of 0.5 / 0.7 plus 0.5 now I'm going to work on the T part so P of this is p of T times I'm going to do log of 0.5 / 0.3 and I do not know what those are off hand uh the first term looks something like 0.336 okay is that the final oh go ahead no that's just the first term I guess I could give you the final it's uh are we missing a a negative sign in front of the first ter no I don't think so um do you see that in the formula I don't see any negatives in the formula oh sorry this is K division I'm just yeah yeah no problem absolutely so the final thing is about 0.59 I believe okay cool um does anyone else agree I mean basically I'm just asking if someone will type this into wol from alpha or oh actually uh that's incorrect that's negative 0.08 I missed a minus sign let's see why is there a minus sign uh math.log 0.5 on 0.7 I see um oh and I missed I missed a 0.5 as well sorry I'm causing all kinds of trouble here it's okay uh 0.5 times that is that and then 0.5 times that is that I'm gonna stop being ask you I'm gonna stop being lazy and we'll both check it positive 0.08 is what I'm getting now I get that get that as well 0.0 um one of the sanity tests we can do is that this number should always be positive and okay so for these two specific coins cack leler Divergence is 0.08 um couple of things to say about the about the K Divergence or the relative entropy one is whenever you calculate it it should be positive or it should at least be um non- negative and it will be zero if and only if if and only if the uh distributions are exactly the same yes yeah so some facts here a l Divergence is I'm just going to call it relative entropy relative entropy is always greater than equal to zero it will be zero if the two distributions are the same and then some interpretations it's a it's like a distance uh it's not a metric in the mathematical sense for one thing it's not symmetric but it is intended to give an idea of how similar two probability distributions are so based on that intuitive interpretation if I picked another biased coin that was closer to being fair um what would you expect to happen with value and I'm about to make a mistake I'm about to say it has a larger or smaller relative inry um it's yeah so I'm about out of time on this session and then I have a a short break that so the the big idea here is uh K Divergence so we have sub KL of P e parallel lines Q is equal to the sum this is true for discret and we replace it with an integral if it's continuous but for now let's do p of x times log P of X over Q ofx so we calculated this where P was a Fair coin and now we want to calculate it and we calculated where P was a Fair coin and Q was a specific bias distribution that had a 70% chance of coming up heads oh okay um so let's go ahead and calculate this bit again uh so this would be 0.5 times log of 0.5 / plus 0.5 * the log of 0.5 / 0.3 this what does log mean is different depending on who you are for a mathematician it's almost always the case that log means natural log log BAS e for an engineer they tend to prefer log base 10 and computer scientists usually often use log base too and it doesn't matter in terms of the qualitative types of information that we get but it makes a big difference in terms of the actual numbers so probably I will have this habit that I need to be careful about of always is writing log for natural log um so the now what we'd like to do is do this more General calculation and so if just qualitatively based on the little bit we know so far about K Divergence if we have a if we had a coin that was more biased um would we expect the KL Divergence to increase or decrease so we have one guess for increase let's see what happens um now we're going to do this with P being the Fair coin and Q being even more biased towards heads so let's say this has a 90% chance of being heads and let calculate so we're going to add up the head is time log of 0.9 and pale is 0.5 * log of 0.5 5 * 0.1 let's say that's equal to I'll calculate this out okay so I got 0.51 I'm using here the natural log um if you use log base 10 you'll get a smaller number but so originally it was smaller than 0.51 so the K Divergence increased so our guess was right and that matches intuitively the fact that K Divergence should be closer to zero for more similar probability distributions and it's actually intended to measure in some sense how similar the two probability distributions are so what we could do is what we did before we could look at this for an arbitrary biased coin with a bias of B and the formula then would be for an arbitrary coin if I just take the same and replace this with b and this with one minus B then I can replace this 0.9 here with b and this 0.1 with one minus B now this is going to be a function of B so we have one number in we can input the bias of our coin towards heads and we can output the KL Divergence from a Fair coin I'm sorry I need to be precise in my terminology we output the K Divergence between the Fair coin and the bias coin and so we can plot this I'm going to instead of using the B I'm going to use the small X my variable name but our function here that depends on one value would be output Y is equal to 0.5 time log of 0.5 / B which is the head's part plus 0.5 time the log of 0.5 / 1 minus B um except I'm going to call this x instead of B just because X and Y are the more common variable names so X is the bias of the coin and Y will be the K Divergence so if I plot that let's see what it looks like kind of interesting plot let's see it so if I look at this part where the bias is the bias of the biased coin is 0.5 it's actually a Fair coin here the K is zero when the coin is actually fair and then what you see is as the coin becomes biased and One Direction or another towards heads or tails the kale Divergence begins to blow up so that K Divergence would actually be approaching Infinity for a totally um for a coin that was totally biased and always landed on H heads or always landed on Tails so this gives us another qualitative tool of understanding uh how the K Divergence depends on the specific distributions that we're looking at now my goal in doing this was to make this more concrete um chapter the first mention of math is in the book not the appendix but the actual book itself jumps from the sum rule for probability it's one of the first things you would learn in probability two k Divergence in the space of about two pages and that's that's not impossible but I think it helps to have this comfort of going through and saying CH Divergence is something that it is actually possible to compute that you could compute successfully correctly and now we can apply this to any probability distribution that we want to look at including uh continuous ones which would just require the definition of entropy and the kale Divergence require passing from a sum to an integral at that point but philosophically it's the same that we are calc using calculations based on the probabilities that we know to determine how similar or dissimilar these two probability distributions are youall have any questions thoughts or comments about that that's all that I had prepared and if I had to do this over I would start with surprise as the negative logarithm of probability and then go to entropy and then go to k Divergence U surprisal it's the negative logarithm of probability do y'all have any questions or thoughts that y'all want to add or share at this point I'm always open to hear what things can we do on our own to strengthen our understanding of what you covered today um I think that doing some calculations I know that people want to skip calculations sometimes and say I'm just focusing on the big picture but often doing the calculations builds belief that you do understand it so perhaps we can create a sequence of notes or exercises that will enable people to practice and see what they understand I think that would be the best um I'm happy to to look at others resources and I think we can also create some of our own the second thing is just keep engaging um it's possible that some people are naturally good at this but the people I've known who were good at it spent time on it and even if you don't intend to do calculation intensive work in the future it can still be like a kind of mental yoga that can support you in other ways do those question do those help at all uh cool great we've actually not done Bas rule but uh the which would be something else that we could do but I'm going by what I see first in the actual text of the book and the first thing it jumps into is jumps pretty quickly into is this K Divergence so getting comfortable with that might be helpful cool well I really appreciate yall coming out um I am afraid I will have to miss next Tuesday I might look at doing next Friday as a makeup but I'd like to uh put together some materials on this and then we can y'all can certainly Aid in that and looking forward to see what happens and appreciate yall coming out all right take care bye